🏦 Bank Peer Analysis PoCA

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Target Stakeholders: Executives, Regulatory Risk.

An interactive data exploration tool built with Streamlit and PyGWalker for analyzing banking institutions' performance and capital metrics using FFIEC data. This PoC was built for financial data analysis and to display rapid prototyping.

🚀 Overview

This application serves as a Proof of Concept for financial analysts to filter, visualize, and export bank peer data. Instead of static charts, it integrates an interactive drag-and-drop visualization engine.

Key Features

- **Interactive Explorer:** Uses PyGWalker to provide a Tableau-like interface for ad-hoc data visualization.
- **Dynamic Filtering:** Sidebar controls to filter institutions by Total Assets (in billions).
- **Excel Export:** Clean data extraction of filtered results using xlsxwriter.
- **Smart Caching:** Optimized data loading for better performance.

🛠️ Tech Stack

- **Frontend:** Streamlit
- **Visualization:** PyGWalkerData
- **Handling:** Pandas & XlsxWriter

📂 Project Structure

```
text├── app.py                  # Main Streamlit application
├── data/
│   └── filtered_bank_data.csv  # Input dataset (FFIEC data)
```

```
├── requirements.txt           # Project dependencies
└── README.md
```

⚙️ Installation & Setup

1. **Clone the repository:**

```
git clone https://github.com
cd bank-peer-analysis
```

2. **Install dependencies:**

```
pip install streamlit pandas pygwalker xlsxwriter
```

3. **Prepare your data:**

   Ensure your bank data is located at data/filtered_bank_data.csv. The script expects a column named Total Assets ($000).

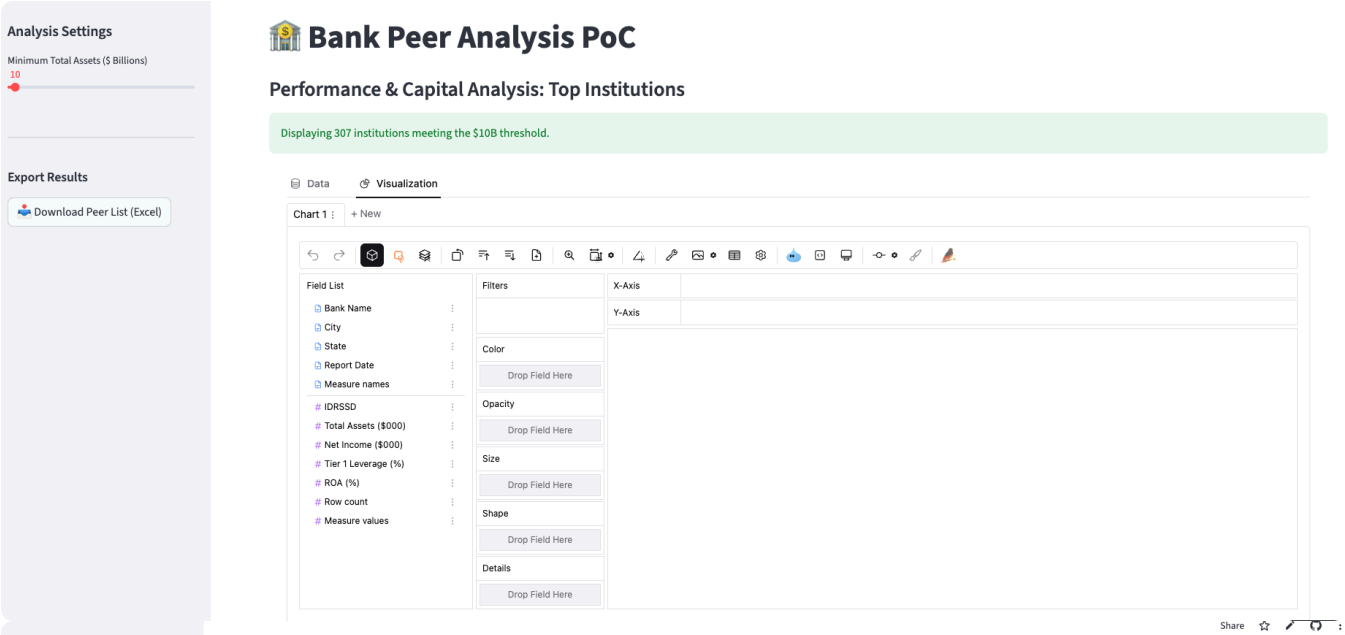
4. **Run the app:**

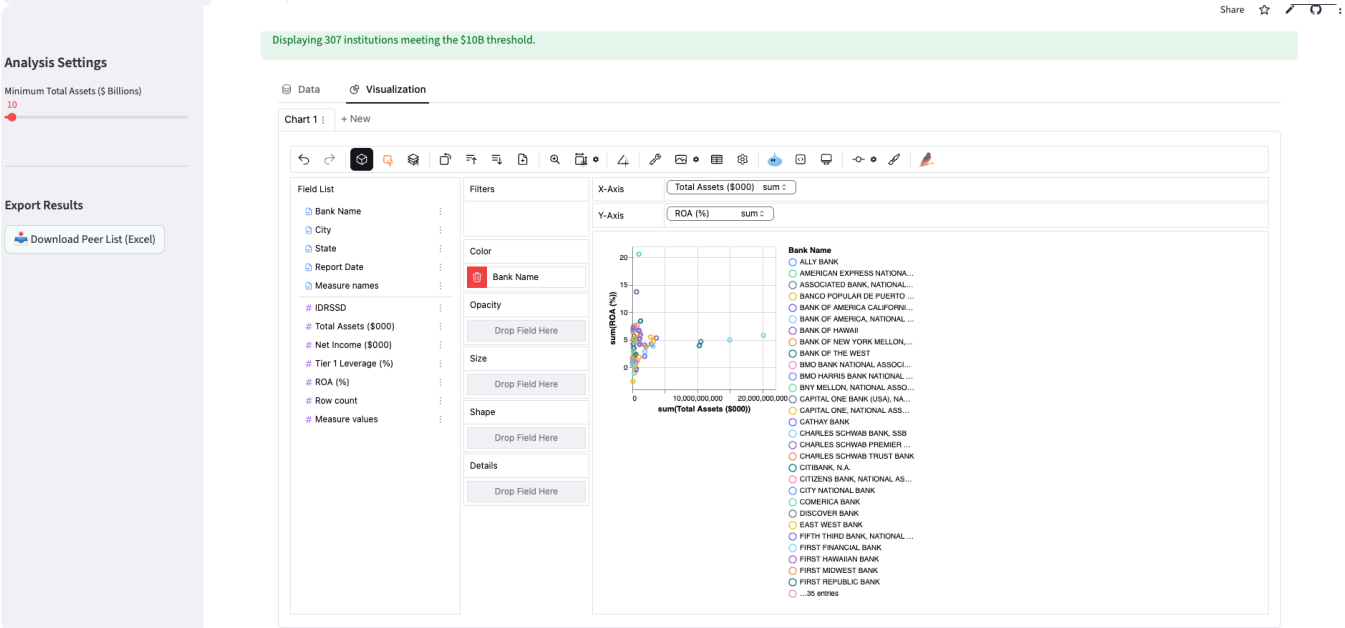
```
streamlit run app.py
```

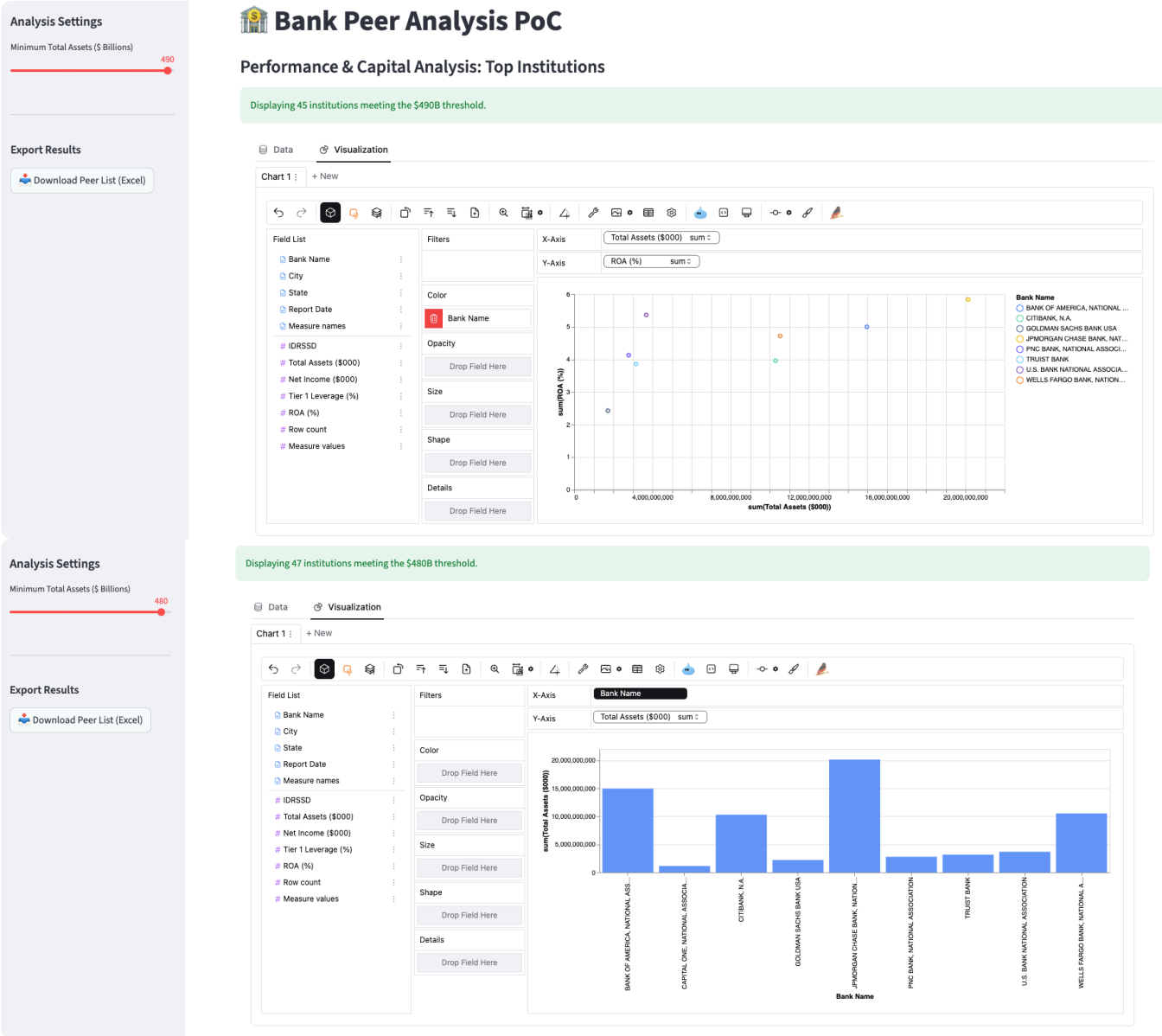
📊 How to Use

1. **Adjust Thresholds:** Use the sidebar slider to set the minimum asset size (e.g., banks with > $50B in assets).
2. **Explore Data:** Use the PyGWalker interface to drag dimensions (Bank Name, State) and measures (Total Assets, ROA) into the shelf to create instant charts.
3. **Export:** Click the Download Peer List button to save your filtered dataset as an Excel file.

Screenshots of PoC






Analysis Settings
Minimum Total Assets ($ Billions)
490
Export Results
Download Peer List (Excel)
Bank Peer Analysis PoC
Performance & Capital Analysis: Top Institutions
Displaying 45 institutions meeting the $490B threshold.
Data
Visualization
Chart 1
+ New
Field List
Bank Name
City
State
Report Date
Measure names
IDRSSD
Total Assets ($000)
Net Income ($000)
Tier 1 Leverage (%)
ROA (%)
Row count
Measure values
Filters
Color
Opacity
Size
Shape
Details
Drop Field Here
X-Axis
Y-Axis
Total Assets ($000) sum
ROA (%) sum
sum(ROA (%))
sum(Total Assets ($000))
Bank Name
BANK OF AMERICA, NATIONAL ...
CITIBANK, N.A.
GOLDMAN SACHS BANK USA
JPMORGAN CHASE BANK, NAT...
PNC BANK, NATIONAL ASSOCI...
TRUIST BANK
U.S. BANK NATIONAL ASSOCIA...
WELLS FARGO BANK, NATION...
Analysis Settings
Minimum Total Assets ($ Billions)
480
Export Results
Download Peer List (Excel)
Displaying 47 institutions meeting the $480B threshold.
Data
Visualization
Chart 1
+ New
Field List
Bank Name
City
State
Report Date
Measure names
IDRSSD
Total Assets ($000)
Net Income ($000)
Tier 1 Leverage (%)
ROA (%)
Row count
Measure values
Filters
Color
Opacity
Size
Shape
Details
Drop Field Here
X-Axis
Y-Axis
Bank Name
Total Assets ($000) sum
sum(Total Assets ($000))
BANK OF AMERICA, NATIONAL ASS...
CAPITAL ONE, NATIONAL ASSOCIA...
CITIBANK, N.A.
GOLDMAN SACHS BANK USA
JPMORGAN CHASE BANK, NATION...
PNC BANK, NATIONAL ASSOCIATION
TRUIST BANK
U.S. BANK NATIONAL ASSOCIATION
WELLS FARGO BANK, NATIONAL A...